TRISOLV — Triangle solver (SSS / SAS / ASA)

TI-84 Evo / Evo-T · Python App
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Pick what you know about a triangle - three sides, two sides and the angle between them, or two angles and the side between them - and the calculator returns all three sides, all three angles in degrees, the area and the perimeter.

WHAT IT ASKS YOU

1. Mode: 1 for SSS, 2 for SAS, 3 for ASA
2. SSS: the three sides a, b, c
3. SAS: sides a and b, then angle C in DEGREES (C sits between a and b)
4. ASA: angles A and B in DEGREES, then side c (c sits between A and B)

WHAT IT SHOWS

- All three sides to 3 decimals, each paired with its opposite angle in degrees
- Area = area of the triangle
- Perim = a + b + c

THE MATH

```
SSS: A = acos((b^2+c^2-a^2) / 2bc)
SAS: c = sqrt(a^2+b^2-2ab*cos C)
ASA: C = 180-A-B, a = c*sin A / sin C
Area (Heron): s=(a+b+c)/2
  area = sqrt(s(s-a)(s-b)(s-c))
```

GET IT ONTO THE CALCULATOR

1. Open **connectevo.ti.com** in Chrome (v143+) and accept the terms.
2. Plug the calculator in with a USB-C *data* cable and switch it on.
3. Click **CONNECT TO CALCULATOR**, pick your calculator, click **Connect**.
4. Click **SEND TO CALCULATOR** and choose `TRISOLV.py`.
5. Check the name reads `TRISOLV`, not `PYTHON01`. Send.

RUN IT

`apps` → **Python App** → highlight `TRISOLV` → **Run**. Answer each prompt and press `enter`.

CHECK IT WORKS

```
TRIANGLE SOLVER
1 SSS  2 SAS  3 ASA
Mode: 1
a = 3
b = 4
c = 5
a = 3.000  A = 36.870
b = 4.000  B = 53.130
c = 5.000  C = 90.000
Area  = 6.000
Perim = 12.000
```

GOOD TO KNOW

- ANGLES ARE ALWAYS IN DEGREES at the prompts. Enter 45, not 0.7854.
- In SAS, angle C is the one BETWEEN sides a and b. In ASA, side c is the one BETWEEN angles A and B. Get this wrong and the answer is wrong, not an error.
- Sides and angles must be positive; 0 or a negative is re-prompted.
- An impossible triangle is caught: sides 1, 2, 10 gives "Not a triangle" rather than a crash.
- In ASA, two angles summing to 180 or more is rejected.

SOURCE CODE — TRISOLV.PY

```
# TRISOLV -- triangle solver: SSS, SAS, ASA
# TI-84 Evo / Evo-T / CE Python / 83 Premium CE Python
#
# Numeric output only. No CAS.
# ANGLES ARE IN DEGREES at every prompt. Python's math module is always
# in radians, so this program converts with radians() internally. The
# calculator's MODE setting does not affect Python -- leave it alone.

from math import *


def getnum(msg):
    while True:
        s = input(msg)
        if "," in s and "(" not in s:
            s = s.replace(",", ".")
        try:
            return float(eval(s))
        except:
            print("Not a number")


def getpos(msg):
    while True:
        v = getnum(msg)
        if v > 0:
            return v
        print("Must be > 0")


def show(a, b, c, A, B, C):
    print("a = {:.3f}  A = {:.3f}".format(a, A))
    print("b = {:.3f}  B = {:.3f}".format(b, B))
    print("c = {:.3f}  C = {:.3f}".format(c, C))
    s = (a + b + c) / 2
    area = sqrt(s * (s - a) * (s - b) * (s - c))
    print("Area  = {:.3f}".format(area))
    print("Perim = {:.3f}".format(a + b + c))


print("TRIANGLE SOLVER")
print("1 SSS  2 SAS  3 ASA")

while True:
    m = int(getnum("Mode: "))
    if m in (1, 2, 3):
        break
    print("Pick 1, 2 or 3")

if m == 1:
    a = getpos("a = ")
    b = getpos("b = ")
    c = getpos("c = ")
    if a + b <= c or a + c <= b or b + c <= a:
        print("Not a triangle")
    else:
        A = degrees(acos((b * b + c * c - a * a) / (2 * b * c)))
        B = degrees(acos((a * a + c * c - b * b) / (2 * a * c)))
        show(a, b, c, A, B, 180 - A - B)

elif m == 2:
    a = getpos("a = ")
    b = getpos("b = ")
    C = getpos("angle C (deg) = ")
    if C >= 180:
        print("Angle must be < 180")
```

- Any prompt accepts an expression or a decimal comma: 4,5 and 9/2 both work.

> Leave the calculator's MODE alone. Python's math module always works in radians, and the MODE screen only governs TI-BASIC and the home screen - it does NOT change Python. TRISOLV takes your degrees and converts them internally with radians(), then converts answers back with degrees().

```
    else:
        c = sqrt(a * a + b * b - 2 * a * b *
cos(radians(C)))
        A = degrees(acos((b * b + c * c - a * a) / (2 * b
* c)))
        show(a, b, c, A, 180 - A - C, C)

else:
    A = getpos("angle A (deg) = ")
    B = getpos("angle B (deg) = ")
    c = getpos("side c = ")
    if A + B >= 180:
        print("A + B must be < 180")
    else:
        C = 180 - A - B
        k = c / sin(radians(C))
        show(k * sin(radians(A)), k * sin(radians(B)), c,
A, B, C)
```